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Research article

Safe Active Discrimination of Fractional, Delayed, and Finite Latent Memory in a Strong-Allee Predator–Prey Model

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Abstract: Can a controlled experiment tell fractional memory apart from the ordinary kinds — integer-order dynamics, delayed feedback, or a few hidden state variables? We pose this question in a strong-Allee predator–prey system whose ecological backbone (model, locked parameters, coexistence equilibrium, and Jacobian) is deliberately inherited from our previously submitted companion paper and is not claimed as new here. Four analytical results organize the study, with an expanded numerical campaign around them. On the structural side, the collocated Caputo transfer carries a noninteger high-frequency signature that no member of the declared finite latent or retarded-delay classes reproduces exactly; on the practical side, finite exponential mixtures track fractional memory over any finite horizon, so discrimination hinges on the complexity granted to the rival. Working directly on the ecological prey channel, interval arithmetic certifies finite-state approximations whose L^1 error falls from 0.5335 at $m = 4$ to 0.0070 at $m = 32$ ($\alpha = 0.85$, $A = 0.25$, $T = 12$); under the declared pulse budget the induced Gaussian testing lower bound climbs essentially to chance. A combined fractional-delay model — reducing to pure Caputo at $\tau = 0$ and to the DDE at $\alpha = 1$ — is then simulated in the interior of the (α, τ) plane, where delay shifts the predator peak, erodes the Allee safety margin, and generates responses that neither pure limit captures. Across six perturbation families and a large four-class benchmark, the pattern is consistent: broadband inputs discriminate better and cross the Allee threshold more often, while transient inputs stay numerically safe and learn less. Experimental identification of ecological memory is thus governed by a genuine safety–informativeness trade-off.

Keywords: fractional-order dynamics; time-delay systems; model discrimination; experimental design; Allee effect; predator–prey dynamics

Mathematics Subject Classification: 34A08; 34K20; 62K05; 92D25

1. Introduction

Fractional derivatives have become a standard modelling device wherever a system appears to remember its past: viscoelastic materials, anomalous transport, and, increasingly, population dynamics [4, 6, 11, 16]. Predator–prey systems in particular have attracted a sizeable fractional-order literature [1, 10, 17, 18]. Yet there is an empirical catch that this literature rarely confronts head-on. Fitting a fractional order α to ecological data does not, by itself, demonstrate that the underlying mechanism is fractional. On any finite observation window, a delayed feedback loop or a handful of unmeasured slow variables can trace out trajectories that look much the same. Methods for designing experiments that estimate a fractional order do exist [8, 9, 13, 20], and safety-aware information gathering has been studied in fair generality [15, 19]. What is missing is the combination, and that is where this paper sits. We ask:

Which perturbations and measurements can distinguish fractional memory from delayed or latent alter-

natives in a strong-Allee ecological system, and when does ecological safety make that discrimination fundamentally difficult?

The word “safety” is doing real work in that sentence. A strong Allee effect means the prey population collapses irreversibly if it is pushed below a threshold, so the very perturbations that would best expose the memory mechanism are also the ones most likely to destroy the system being studied.

Our starting point is an ecological model we did not build here. The backbone — the strong-Allee predator–prey equations, their locked parameters, the coexistence equilibrium, and its Jacobian — comes from a companion study [3, 5, 12, 21] in which those objects were certified with validated numerics. Reusing them is a deliberate choice: it lets the present paper concentrate on one question, experimental discrimination of memory mechanisms, without re-litigating the dynamics. Nothing from the companion paper is claimed as new. What *is* new here are the rival response laws themselves (integer-order, Caputo, retarded-delay, and finite latent), the controlled perturbations and observation channels through which they are compared, the finite-horizon discrimination bounds, a combined fractional-delay stress test, and the experiment-design benchmark. Only the integer-order and Caputo mechanisms share the inherited equilibrium and Jacobian exactly; the delay and latent rivals are calibrated alternative response laws, matched to the same operating point, input channels, and physical units.

Explicit relationship to the previously submitted paper. Since the two manuscripts form a sequence, we prefer to draw the reuse boundary precisely rather than leave a reviewer guessing. The inherited ingredients are exactly four: the strong-Allee equations, the locked ecological parameters, the coexistence equilibrium, and the Jacobian at that equilibrium, all taken from [12]. None of them re-enters the novelty claim. They supply a controlled test bed — one whose dynamical regimes are already certified — on which a different problem can be posed: how should one perturb and observe such a system to tell candidate memory laws apart? Table 1 spells out the division of labour.

Table 1. Relationship between the previously submitted companion paper and the present manuscript. The ecological backbone is intentionally shared; the research questions, analytical targets, simulations, and headline results are different.

Aspect	Previously submitted paper [12]	Present paper
Scientific question	Certify equilibria, stability, the Matignon boundary, extinction/boundedness, and discretization-induced stability inversion.	Design and evaluate experiments that distinguish Caputo, delayed, integer-order, and finite-latent memory mechanisms.
Shared material	Strong-Allee ecological equations, locked parameters, coexistence equilibrium, and Jacobian.	The same ecological backbone is reused only as the operating and safety test bed.
Primary mathematics	Validated numerics, Krawczyk/interval certificates, Matignon classification, global extinction/boundedness statements.	Structural transfer separation, finite-horizon latent approximation, testing obstruction, and finite-state prey-response approximation.
Primary computation	Certified parameter grid and certificate inventory.	Frequency-response diagnostics, input-design experiments, four-class model-discrimination benchmark, and fractional-delay simulations.
Headline output	Certified dynamical regimes of the Caputo ecological model.	Safety–informativeness limits for active discrimination of competing memory mechanisms.

There is also a methodological reason to keep the two papers apart. A stability certificate tells you whether a given Caputo model is consistent with an observed dynamical regime. It says nothing about whether that Caputo model could be told apart from a delay or latent-state mechanism producing nearly the same observations — a different question with a different mathematical shape. In the other direction, nothing below recertifies the backbone; we simply condition on the established operating geometry and ask how much extra perturbation and measurement information the discrimination task requires.

Main contributions. Rather than presenting a long chain of standalone theorems, we organise the paper around four contributions.

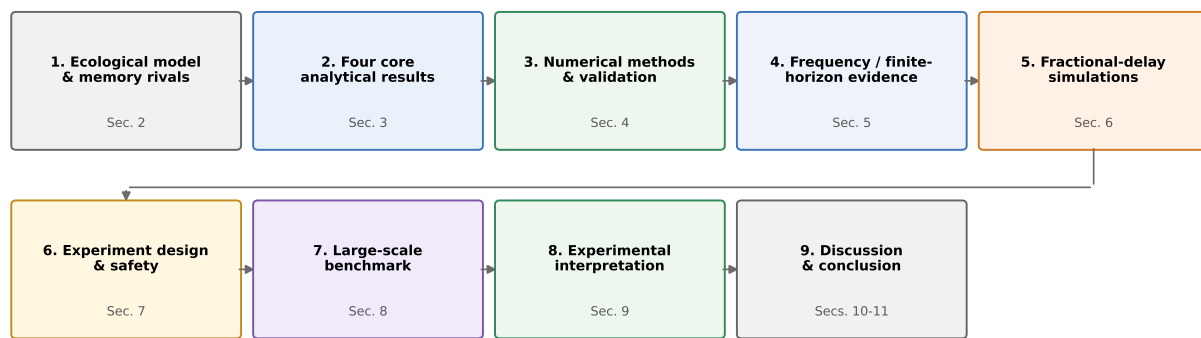
- 1. Structural and finite-horizon separation.** The collocated Caputo transfer carries a noninteger high-frequency signature that no finite rational latent model, and no member of the declared retarded-delay class, can reproduce exactly. At the same time, finite exponential mixtures do approximate the fractional kernel on any fixed horizon — so the ability to discriminate depends on the complexity one is willing to grant the rival, and that dependence can be quantified.
- 2. Certified prey-response testing obstruction.** Working directly on the linearized ecological prey channel, a pole/branch decomposition produces finite-state approximations whose L^1 errors are enclosed by interval

arithmetic. Feeding those enclosures into a Gaussian observation model under the declared pulse budget yields a minimax testing lower bound that climbs toward chance as the latent complexity grows.

3. **Numerical validation and a fractional-delay stress test.** Alongside the calibrated four-class benchmark (integer-order, Caputo, delayed, latent), we simulate a combined fractional-delay model whose two boundaries are exactly the mechanisms previously studied in isolation: $\tau = 0$ recovers the pure Caputo model and $\alpha = 1$ recovers the DDE. This makes it possible to watch fractional order and delay jointly reshape phase, transient amplitude, safety margin, and discriminability.
4. **Safety-informativeness trade-off.** Across six input families the pattern is consistent and, in hindsight, unwelcome: broadband excitation discriminates best and crosses the Allee threshold most often, while transient inputs stay safe and learn little. The constrained design problem is not a theoretical curiosity; it is visible in every experiment we ran.

Detailed proofs, secondary analytical results, the Bayesian sequential extension, and solver derivations are collected in the Supplementary Material. The main text keeps only the analytical statements needed to read the numerical results.

Paper architecture: compact theory, expanded simulations, and a dedicated fractional-delay study



Theory is compacted in the main text; detailed proofs and the Bayesian extension are in the Supplement.

Figure 1. The paper at a glance: certified ecological model and rival memory laws, four essential analytical results, numerical validation, dedicated fractional-delay simulations, experiment design and safety analysis, and the large-scale discrimination benchmark. Detailed proofs and secondary theory appear in the Supplementary Material.

2. Ecological Model and Competing Memory Mechanisms

2.1. Strong-Allee predator-prey backbone

Everything in this subsection is inherited. The model, its locked parameterization, the coexistence equilibrium, the Jacobian, and the Matignon stability boundary were derived and certified in the companion study [12]; we restate the formulas simply because the discrimination experiments cannot be defined without them, and we claim none of them as new.

The nonlinear ecological model reads

$$f_1(x, y) = rx \left(1 - \frac{x}{K}\right) \left(\frac{x}{A} - 1\right) - \frac{axy}{1 + hx}, \quad (2.1)$$

$$f_2(x, y) = e \frac{axy}{1 + hx} - my, \quad (2.2)$$

with locked parameters

$$r = \frac{3}{2}, \quad K = 1, \quad a = 1, \quad h = \frac{1}{2}, \quad e = \frac{4}{5}, \quad m = \frac{2}{5}.$$

The coexistence equilibrium and Jacobian are

$$x^* = \frac{2}{3}, \quad y^*(A) = \frac{2(2 - 3A)}{9A},$$

$$J(A) = \begin{pmatrix} \frac{7A-2}{10A} & -\frac{1}{2} \\ \frac{8A}{2-3A} & 0 \end{pmatrix}.$$

Local stability of the commensurate Caputo model is governed by Matignon's sector condition [14]. In the complex-pair regime $2/7 < A < 10/19$ the critical order works out to

$$\alpha^*(A) = \frac{2}{\pi} \arctan\left(\frac{\sqrt{4D(A) - T(A)^2}}{T(A)}\right),$$

where $T(A) = \text{tr } J$ and $D(A) = \det J$. The number is worth pausing over at one parameter value: at $A = 0.30$ one finds $\alpha^* \approx 0.969012276$, so a fractional system with $\alpha = 0.9$ sits comfortably in the stable regime even though its integer-order limit is unstable [12] — memory alone changes the verdict. Most of our discrimination experiments, however, use $A \leq 2/7$, where the operating point is stable for every candidate mechanism and stability itself cannot be used as a shortcut for telling them apart.

Inherited backbone versus new experimental layer. We kept the notation identical to the companion paper on purpose. Had we altered the ecological equations, any observed difference between memory laws could be blamed on the changed biology rather than on the memory itself. So the ecology stays fixed, and what varies is the *response law wrapped around it*: fractional derivative, explicit delay, or a finite latent state, each determining how a perturbation echoes through time. Every candidate then faces the same perturbation amplitude, the same observation horizon, and the same Allee threshold. Table 2 lists the quantities that are frozen unless a sweep explicitly varies A or α .

Table 2. Locked ecological backbone used by both manuscripts. These values are inherited inputs, not new contributions of the present paper.

Quantity	Value	Role in present experiments
r	1.5	intrinsic prey growth
K	1	prey carrying capacity
a	1	attack-rate scale
h	0.5	Holling-II handling parameter
e	0.8	conversion efficiency
m	0.4	predator mortality
x^*	$2/3$	common coexistence prey level
$y^*(A)$	$2(2-3A)/(9A)$	coexistence predator level
$J(A)$	Eq. above	ODE/Caputo local ecological generator

The Allee threshold enters the story twice. It defines the failure event — push prey below A and the experiment may exit the coexistence regime altogether, with extinction dynamics taking over — and it hands us a natural safety margin in the distance $x^* - A$. An input can therefore be statistically excellent and ecologically indefensible at the same time. That tension is not hypothetical; it shows up repeatedly in the numerical work of Sections 6, 7, and 8.

2.2. Four rival mechanisms

With the reference time τ_0 enforcing dimensional consistency [6, 11, 16], the four candidate mechanisms around the operating point are

$$\begin{aligned} \text{ODE: } \dot{\xi} &= J\xi + Bu, \\ \text{Caputo: } \tau_0^{\alpha-1} D_t^\alpha \xi &= J\xi + Bu, \\ \text{DDE: } \dot{\xi} &= A_0\xi(t) + A_1\xi(t-\tau) + Bu, \\ \text{latent: } \dot{q} &= \bar{A}q + \bar{B}u, \quad y = \bar{C}q. \end{aligned}$$

The Caputo mechanism is the only one carrying a fractional order. The latent class, which would otherwise be flexible enough to imitate anything, is held to an explicit state-dimension budget m — a restriction whose necessity is precisely the content of Result II below.

System, observation channels, and rival memory laws

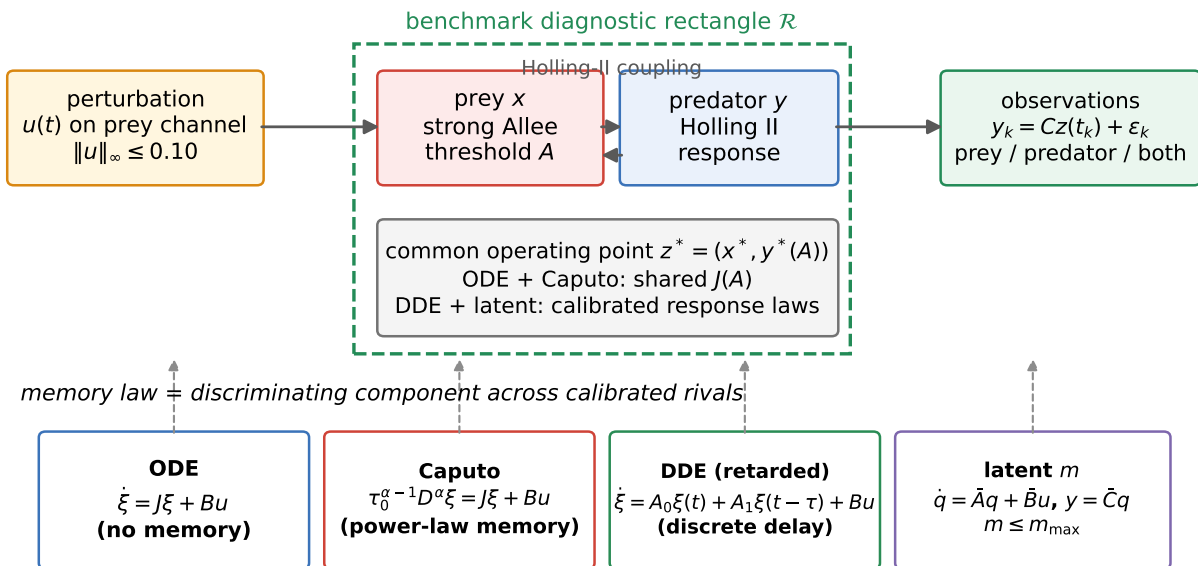


Figure 2. Ecological and experimental setting. The strong-Allee backbone and its parameterization come from the previous validated study [12]; the intervention/observation architecture and the competing memory laws belong to the present work. Perturbations enter on the prey channel; observations may cover prey, predator, or both. ODE and Caputo models share the exact ecological Jacobian, while DDE and latent models are calibrated rival response laws. The dashed rectangle is a numerical benchmark diagnostic region, not an invariant-set certificate.

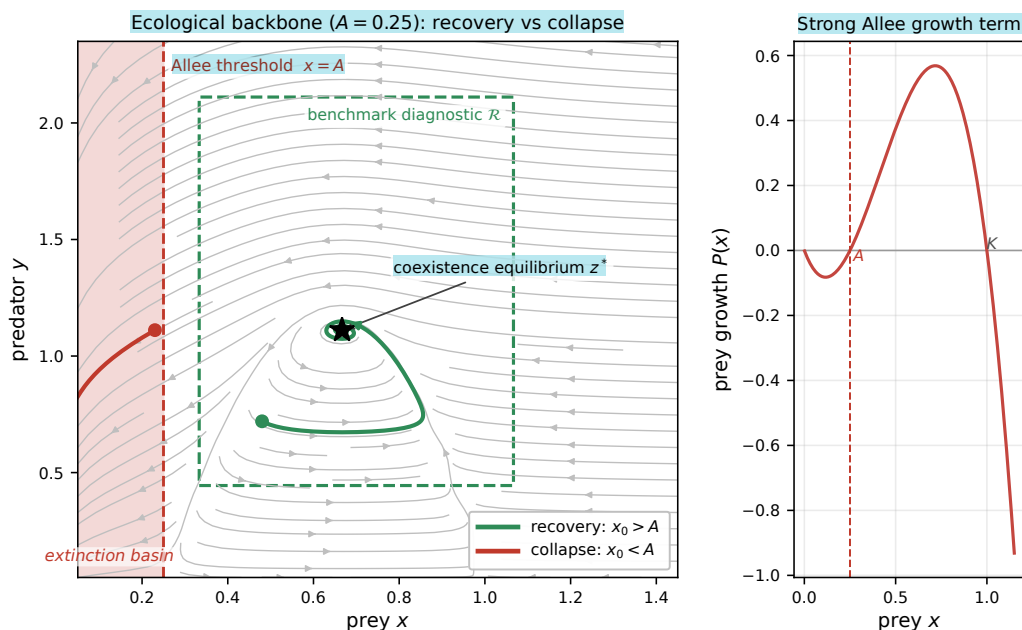


Figure 3. The inherited strong-Allee backbone [12] at $A = 0.25$, shown only to fix the safety geometry for the experiment-design problem. Left: recovery from above the threshold versus collapse from below it. Right: the prey growth term $P(x)$, changing sign at the Allee threshold.

Combined fractional-delay stress-test model. Fractional memory and explicit delay need not be rivals; a real ecosystem could harbour both. To probe their interaction we also simulate

$$\tau_0^{\alpha-1} D_t^\alpha x(t) = P(x(t); A) - \frac{ax(t)y(t)}{1 + hx(t)} + u(t), \quad (2.3)$$

$$\tau_0^{\alpha-1} D_t^\alpha y(t) = e^{\frac{ax(t-\tau)y(t)}{1 + hx(t-\tau)}} - my(t), \quad (2.4)$$

with a constant equilibrium history on $[-\tau, 0]$. Two limits anchor the interpretation: $\tau = 0$ collapses (2.4) onto the pure Caputo model, and $\alpha = 1$ onto the retarded DDE benchmark. We use it strictly as a numerical stress test — no theorem in this paper depends on it.

3. Analytical Backbone: Four Results Used by the Experiments

Four analytical statements carry the weight of the numerical study; everything else — full derivations, secondary theorems, and the Bayesian machinery — lives in the Supplementary Material. We present the four in the order a reader needs them, each with a short account of why it matters.

3.1. Result I: structural separation

On the collocated channel ($CB \neq 0$) the Caputo transfer is

$$G_\alpha(s) = C(q_\alpha(s)I - J)^{-1}B, \quad q_\alpha(s) = \tau_0^{-1}(s\tau_0)^\alpha,$$

with high-frequency expansion

$$G_\alpha(s) = (CB)\tau_0^{1-\alpha}s^{-\alpha} + O(|s|^{-2\alpha}).$$

Theorem 3.1 (Structural separation). *For $0 < \alpha < 1$ and $CB \neq 0$, the Caputo transfer cannot equal any finite-dimensional rational latent transfer on a nonempty open subset of a common analytic domain. It also cannot equal a strictly proper finite-state retarded model with finitely many discrete delays, because the latter has $O(\omega^{-1})$ high-frequency decay whereas the Caputo channel has $O(\omega^{-\alpha})$ decay.*

In other words, the fractional signature is genuinely there: no rival in either declared class reproduces it exactly. What the theorem does *not* say is just as important — it promises nothing about finite, noisy experiments, which is where the next result bites.

3.2. Result II: finite-horizon latent approximation

The Caputo memory kernel admits a diffusive representation, and on a bounded window $[0, T]$ it can be tracked by a positive exponential sum. A constructive m -mode approximation k_m satisfies

$$\|k_\alpha - k_m\|_{L^1(0,T)} \leq \widehat{E}_m, \quad \widehat{E}_m \downarrow 0$$

along an expanding nested rate hierarchy.

Theorem 3.2 (Finite-horizon approximation and complexity barrier). *For every $T < \infty$ and $\varepsilon > 0$, there exists a finite positive exponential mixture whose $L^1(0, T)$ distance from the Caputo kernel is below ε . Consequently, without a finite complexity restriction on the latent rival, no fixed finite-horizon experiment can guarantee a uniform positive separation from the entire latent hierarchy.*

Taken together, Results I and II frame the whole problem. Exact separation holds, yet an unconstrained latent rival can forge the fractional kernel to any tolerance on any finite window. Discrimination is therefore meaningful only relative to a declared complexity budget — which is why m appears throughout this paper as an experimental design parameter rather than a nuisance to be estimated away.

3.3. Result III: testing obstruction under a physical input budget

Write S for the observation operator, R for the Gaussian noise covariance, and set

$$C_{\text{obs}} = \frac{1}{2} \|R^{-1/2}S\|^2.$$

Young's inequality carries the kernel approximation error into observation space, and a two-point Gaussian argument does the rest.

Theorem 3.3 (Complexity-dependent testing obstruction). *If the admissible experiment class satisfies $\|u\|_2 \leq B$, then the composite minimax error for testing the fractional response against a latent hierarchy containing an explicit competitor with error $\widehat{E}_m$ obeys*

$$P_e^*(m) \geq \Psi\left(C_{\text{obs}} \widehat{E}_m^2 B^2\right),$$

where Ψ is any valid equal-covariance Gaussian two-point lower bound (Pinsker is used in the numerical plots). Thus $P_e^*(m) \rightarrow 1/2$ whenever $\widehat{E}_m \rightarrow 0$ and B remains bounded.

The bound has a blunt reading: with a capped input budget, a rich enough rival forces any test toward coin-flipping, no matter how cleverly the experiment is designed.

3.4. Result IV: ecological prey-response approximation

The three results so far concern the abstract memory kernel. The fourth works directly on the ecological quantity an experimenter actually measures: the prey-to-prey fractional transfer. One technical trap deserves mention — the tempting factorization $(s^\alpha I - J)^{-1} = (sI - J)^{-1} s^{-\alpha}$ is simply false, and we avoid it. Instead, a pole/branch decomposition realizes the principal-sheet pole pair *exactly* with a stable two-state integer-order block, so that only the smooth branch remainder needs approximating.

Theorem 3.4 (Certified finite-state approximation of the prey response). *For the displayed Matignon-stable cells and $m \in \{4, 8, 16, 32\}$, there exists an explicit stable finite-state prey-response surrogate g_m such that*

$$\|g_\alpha - g_m\|_{L^1(0,T)} \leq \widehat{E}_m^{\text{state}},$$

where the bound is obtained by outward-rounded interval subdivision and a rigorous tail estimate. At $\alpha = 0.85$, $T = 12$, the certified enclosure at $A = 0.25$ decreases from 0.5335 ($m = 4$) to 0.0070 ($m = 32$).

These are certified enclosures, not floating-point estimates: every displayed constant survives outward rounding. For the declared pulse, $\|u\|_\infty \leq 0.10$ gives $\|u\|_2 \leq 0.120$, and at $\sigma = 0.10$ the resulting Pinsker lower bounds at $A = 0.25$ come out near 0.340, 0.462, 0.492, and 0.498 for $m = 4, 8, 16$, and 32. Read down that list once more: by $m = 32$ the surrogate is, for all practical purposes, statistically invisible against the true fractional response.

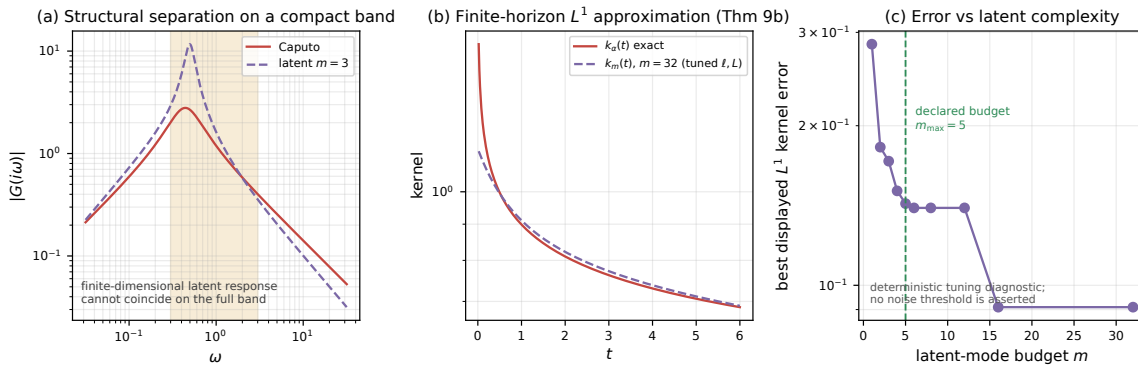


Figure 4. Structural separation and finite-horizon approximation coexist without contradiction. The Caputo transfer carries a distinct noninteger signature, and yet sufficiently rich finite exponential mixtures shadow the fractional kernel over any finite horizon.

4. Numerical Methods and Validation

Numerics in this paper play a supporting role: they illustrate and stress-test the analytical claims, never substitute for them. Every script and every machine-readable output behind the figures is released in the accompanying benchmark and figure directories, so the reader who distrusts a number can regenerate it.

4.1. Solvers

Nothing exotic is used where nothing exotic is needed. ODE and latent-state systems run under classical fourth-order Runge–Kutta; retarded DDEs use an interpolated-history RK4 variant. The Caputo systems require more care and use

Certified prey-response finite-state approximation and testing bound ($\alpha = 0.85$, $T = 12$)

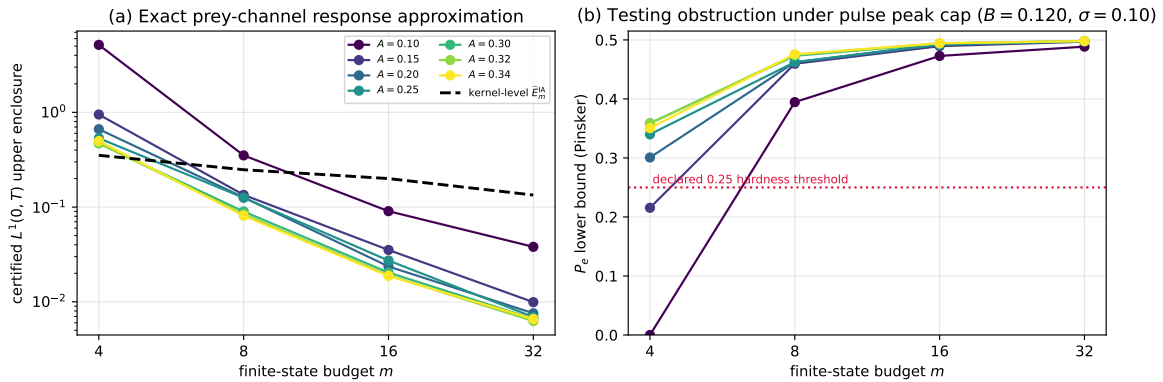


Figure 5. Certified prey-response approximation and the testing obstruction it induces. As the finite-state budget grows, the interval-enclosed response error collapses and the lower bound on testing error climbs toward chance.

the fractional Adams–Bashforth–Moulton PECE scheme [16], whose scalar implementation we validate against the exact Mittag–Leffler solution.

The combined fractional-delay model (2.4) reuses the same PECE history convolution, with the delayed prey value obtained by interpolating the trajectory already computed. All reported grids with $\tau > 0$ satisfy $\tau \geq h$, which guarantees that the corrector only ever asks for history that is already known. The two boundaries of the model double as validation gates:

- at $\tau = 0$ the code delegates outright to the validated Caputo solver;
- at $\alpha = 1$ its output must agree with the independent DDE RK4 path — and it does.

Concretely, at $A = 0.25$ and $\tau = 0.35$ the maximum discrepancy between the $\alpha = 1$ fractional-delay run and the DDE solver falls from 6.75×10^{-6} at $N = 200$ to 1.68×10^{-6} at $N = 400$ and 4.20×10^{-7} at $N = 800$. Away from the boundaries, at the representative cell $(\alpha, \tau) = (0.85, 0.35)$, self-convergence against the $N = 800$ solution improves from 2.18×10^{-3} ($N = 200$) to 8.05×10^{-4} ($N = 400$).

4.2. Inputs, observations, and metrics

Six bounded input families compete throughout, all held to the same peak budget $\|u\|_\infty \leq 0.10$: a single pulse, a multiscale pulse train, a sinusoid, a multisine, a chirp, and a pseudorandom binary sequence (PRBS). **Observations** may cover prey, predator, or both channels. We keep numerical safety separate from mere solver survival: the quantities reported are the minimum Allee margin

$$\min_t (x(t) - A)$$

and the fraction of runs that cross the threshold.

The fractional-delay experiments carry four extra diagnostics: trajectory distance to the pure Caputo limit, trajectory distance to the pure DDE limit, the timing and amplitude of the predator peak, and a bridge-discrimination score defined as the smaller of the two limit distances — small when either pure mechanism explains the data, large only when neither does.

4.3. Validation matrix and numerical resolution

Rather than resting on a single fortunate solver run, each numerical claim is tied to an explicit convergence gate; Table 3 collects them. The scalar Caputo implementation is held against the exact Mittag–Leffler solution, the fractional-delay code is checked at its integer-order boundary against the independent DDE solver, and an interior cell undergoes self-convergence. Every reported error sequence decreases monotonically.

A word on why the gates are heterogeneous. Two implementations that share a discretization can agree beautifully while both being wrong, so wherever possible we compare against a mathematically different reference. At $\tau = 0$ duplication would prove nothing, and the code simply hands over to the already validated Caputo path; at $\alpha = 1$, where the memory convolution degenerates to an integer-order problem, the comparison runs instead against the separate interpolated-history DDE solver. Between them, the two checks pin down opposite boundaries of the combined model.

Table 3. Numerical validation gates. The Caputo column uses the exact Mittag–Leffler reference; the fractional-delay boundary uses an independent DDE RK4 solution.

Gate	grid size	Reference	max. abs. discrepancy
Caputo PECE	$N = 250$	exact Mittag–Leffler	8.70×10^{-4}
Caputo PECE	$N = 500$	exact Mittag–Leffler	4.64×10^{-4}
Caputo PECE	$N = 1000$	exact Mittag–Leffler	2.52×10^{-4}
Fractional-delay, $\alpha = 1$	$N = 200$	DDE RK4	6.75×10^{-6}
Fractional-delay, $\alpha = 1$	$N = 400$	DDE RK4	1.68×10^{-6}
Fractional-delay, $\alpha = 1$	$N = 800$	DDE RK4	4.20×10^{-7}
Fractional-delay, $(\alpha, \tau) = (0.85, 0.35)$	$N = 200$	$N = 800$	2.18×10^{-3}
Fractional-delay, $(\alpha, \tau) = (0.85, 0.35)$	$N = 400$	$N = 800$	8.05×10^{-4}

4.4. Experimental grid and reproducibility

The benchmark is built on a handful of interpretable axes, not on open-ended parameter search. The peak actuator budget $\|u\|_\infty \leq 0.10$ applies everywhere. The four-class stable benchmark sweeps Allee level, fractional order, SNR, observation channel, and waveform family at 200 replicates per cell; the fractional-delay campaign is much smaller and deliberately so, using $A = 0.25$ with $\alpha \in \{0.75, 0.85, 0.95\}$ and $\tau \in \{0.15, 0.30, 0.45\}$ for the design comparison plus a denser (α, τ) grid for the bridge plots. Table 4 summarizes.

Table 4. Main numerical-design axes. The fractional-delay campaign is intentionally targeted and is separate from the larger four-class benchmark.

Component	Values / protocol
Peak input budget	$\ u\ _\infty \leq 0.10$ for all six waveform families
Four-class stable benchmark	ODE, Caputo, retarded DDE, latent-3; $A \in \{0.20, 0.25\}$; 200 replicates per cell
Fractional orders in benchmark	$\alpha \in \{0.70, 0.85, 0.90, 0.95\}$ where applicable
Observation channels	prey only, predator only, both species
Noise regimes	high, medium, and low SNR
Fractional-delay design grid	$A = 0.25$, $\alpha \in \{0.75, 0.85, 0.95\}$, $\tau \in \{0.15, 0.30, 0.45\}$
Fractional-delay bridge grid	$\alpha = 0.70, 0.75, \dots, 1.00$ and $\tau = 0, 0.10, \dots, 0.50$
Simulation horizon	$T = 12$ for the principal trajectory/design comparisons

Every figure quantity traces back to a JSON or NPZ file written by the simulation scripts; the fractional-delay script in particular emits its validation sequence, design ranking, and bridge arrays alongside the four figures it draws. Keeping simulation output separate from paper rendering pays off at audit time — plots can be redrawn without touching the archived four-class benchmark, and the archived statistics can be re-examined without redrawing anything.

5. Frequency-Domain and Finite-Horizon Evidence

No large benchmark is needed to see the structural results at work; two frequency-domain plots suffice. Figure 6 compares transfer magnitudes on the prey channel at $A = 0.25$, and Figure 7 tracks the phase, whose collocated-channel limit $-\alpha\pi/2$ is the fractional fingerprint in its purest form.

5.1. From exact signatures to finite-horizon ambiguity

The two panels reward a joint reading. Asymptotically, the noninteger decay and phase brand the Caputo channel unmistakably — but no experiment observes an asymptote. **What an experiment sees is a slice of bandwidth, over a window of time, through a veil of noise, and on that slice a flexible latent model can mimic the relevant portion of the response while remaining analytically distinct from it. Far from contradicting each other, the structural theorem and the finite-horizon impossibility theorem describe the two sides of this same coin.**

The prey-response calculation brings the point home in the channel an ecologist would actually instrument. Table 5 lists, for $\alpha = 0.85$ and $T = 12$, the certified L^1 response bounds and the Pinsker lower bounds they induce under the declared pulse at $\sigma = 0.10$. Between $m = 4$ and $m = 32$ the approximation tightens by nearly two orders of magnitude, and the testing bound duly climbs to within a hair of the equal-prior chance level $1/2$.

One caution about reading the table: it is a one-sided instrument. A large lower bound on P_e^* proves the specified experiment cannot reliably separate the fractional response from at least one admitted finite-state rival. A small bound proves nothing in the other direction — such cells are inconclusive, not certified-easy. The atlas of Figure 8 inherits

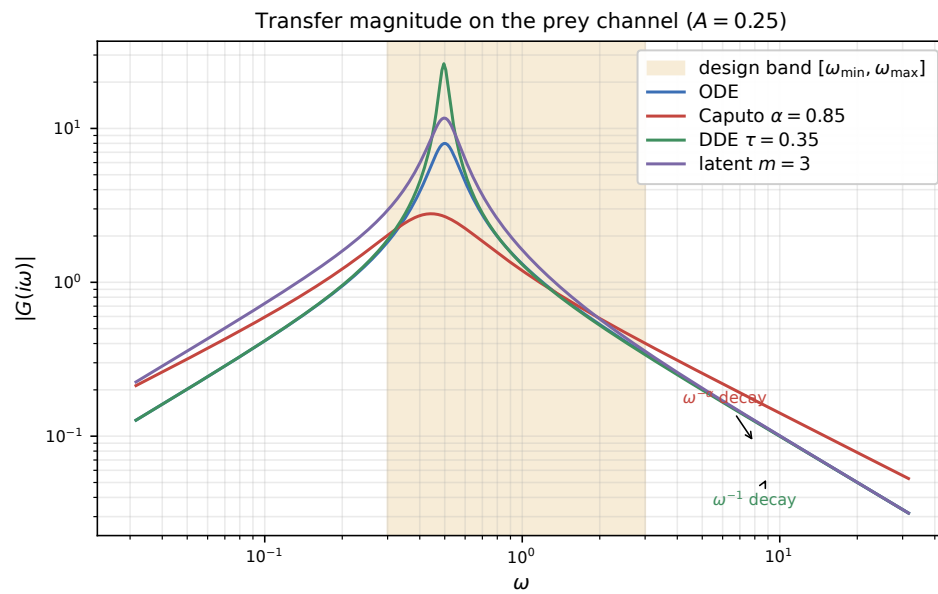


Figure 6. Transfer magnitude on the prey channel for the four memory mechanisms at $A = 0.25$. The fractional response separates across the design band and settles onto its noninteger high-frequency decay.

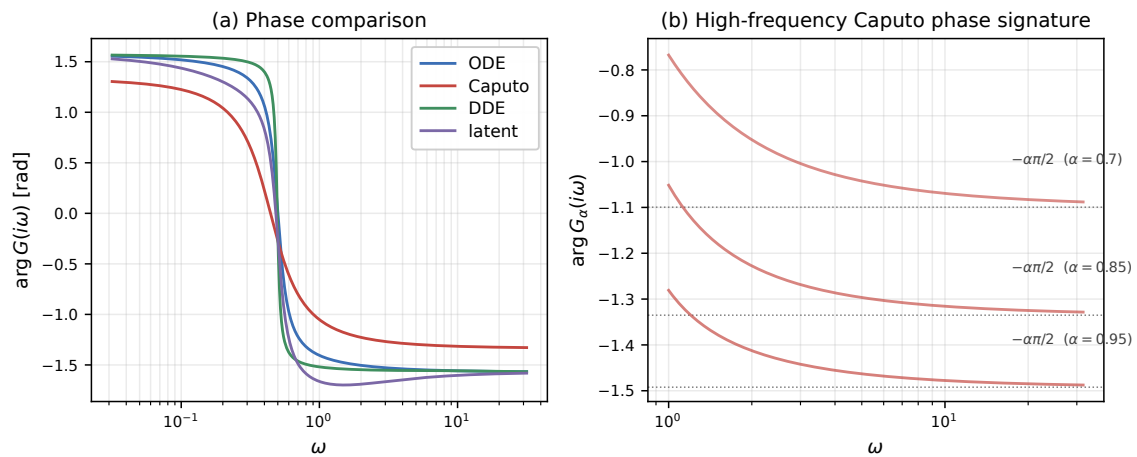


Figure 7. Phase evidence for the fractional signature. Left: Caputo, ODE, DDE, and finite latent responses across frequency. Right: the Caputo phase converging to the scale-free asymptote $-\alpha\pi/2$.

Table 5. Certified finite-state approximation of the fractional prey response and induced testing lower bound at $\alpha = 0.85$, $T = 12$, pulse $\|u\|_2 \leq 0.120$, and $\sigma = 0.10$.

m	$A = 0.25$		$A = 0.30$	
	$\hat{E}_m^{\text{state}}$	P_e^* lower bound	$\hat{E}_m^{\text{state}}$	P_e^* lower bound
4	0.5334	0.340	0.4706	0.359
8	0.1254	0.462	0.0898	0.473
16	0.0273	0.492	0.0204	0.494
32	0.0069	0.498	0.0066	0.498

exactly this reading: it maps where discrimination is provably hard across fractional order, Allee parameter, and latent complexity, with cells outside the focal Matignon-stable regime hatched out of the headline interpretation.

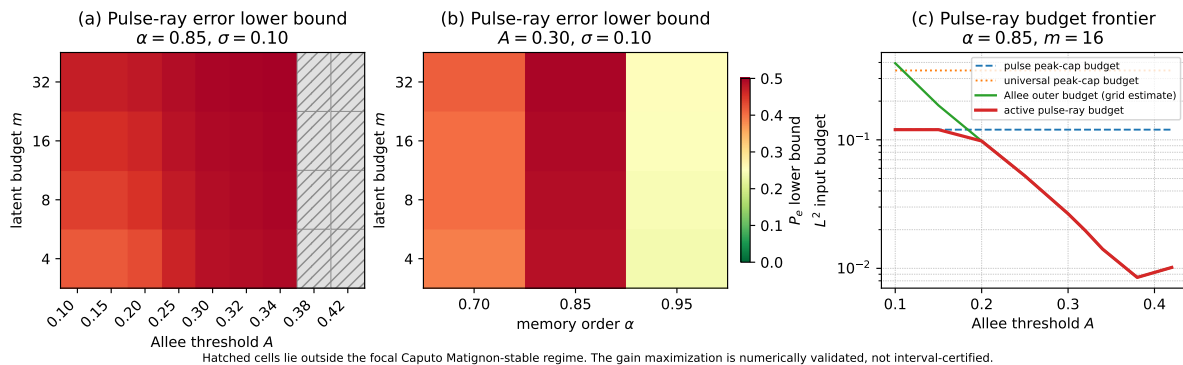


Figure 8. Protocol-restricted memory-discrimination atlas. Growing latent complexity, or drifting toward weakly separated fractional regimes, raises the lower bound on model-selection error. Hatched cells fall outside the focal Caputo stable regime.

6. Fractional-Delay Model: Dedicated Simulation Study

So far, fractional memory and explicit delay have been treated as competing explanations. Nature is under no obligation to choose between them. This section studies a mechanism carrying *both*, using the combined model (2.4), whose two boundaries are precisely the mechanisms already characterized in isolation: setting $\tau = 0$ recovers the pure Caputo model, setting $\alpha = 1$ recovers the DDE benchmark. Whatever happens in the interior of the (α, τ) plane tells us whether the two effects reinforce each other, cancel, or produce responses that neither pure explanation can absorb.

We want to be clear about ambition. This is not an attempt at a general theory of fractional delay equations; it is an application-oriented stress test of the discrimination framework, run on the same strong-Allee ecology, the same prey perturbation channel, the same amplitude budget, and the same observation variables as everything else in the paper. Only the two memory coordinates move. The payoff is a view of something pairwise comparisons hide: if a real system harbours both a broad hereditary response and a physiologically meaningful lag, then forcing a binary “fractional versus delay” verdict misspecifies the problem even when each limiting model fits part of the data.

6.1. Transient trajectories

Figure 9 holds $A = 0.25$ and $\alpha = 0.85$ fixed, applies one and the same prey pulse, and lets τ grow. The prey barely notices. The predator does: its response shifts visibly, and the late transient becomes more and more delay-sensitive. For this pulse experiment, every trajectory stays above the Allee threshold.

6.2. Fractional-order/delay parameter plane

Figure 10 sweeps $\alpha \in [0.70, 1]$ against $\tau \in [0, 0.5]$. By construction the left edge sits at zero distance from the pure Caputo model and the top edge at zero distance from the pure DDE; the interesting question is how fast the interior escapes both. The answer, per the safety panel, comes with a warning: a delay that is harmless at low α eats into the Allee margin as the system approaches the integer-order limit.

Table 6 pulls representative numbers off the grid. At fixed α , growing τ pushes the trajectory monotonically away from the Caputo boundary while it stays well clear of the DDE boundary in every displayed cell. Near $\alpha = 0.95$ the effect sharpens considerably: the Caputo distance climbs from 0.0299 at $\tau = 0.10$ to 0.2805 at $\tau = 0.50$, and the minimum Allee margin drops from 0.3345 to 0.2629 over the same range. Here, in one concrete corner of parameter space, explicit delay buys identifiability and ecological risk in the same purchase.

6.3. Which input exposes the combined mechanism?

To rank inputs, we score each one over $\alpha \in \{0.75, 0.85, 0.95\}$ and $\tau \in \{0.15, 0.30, 0.45\}$ by the mean of its minimum trajectory distance to the two limiting explanations — a score built to punish inputs that merely separate the combined response from *one* of the limits while leaving the other plausible. The outcome (Figure 11, Table 7) has a familiar shape. Multisine excitation wins on discrimination and pays for it, crossing the Allee threshold in a third of the tested cells. PRBS, sinusoid, and chirp cross in every single cell. The two transient families, pulse and multiscale, never

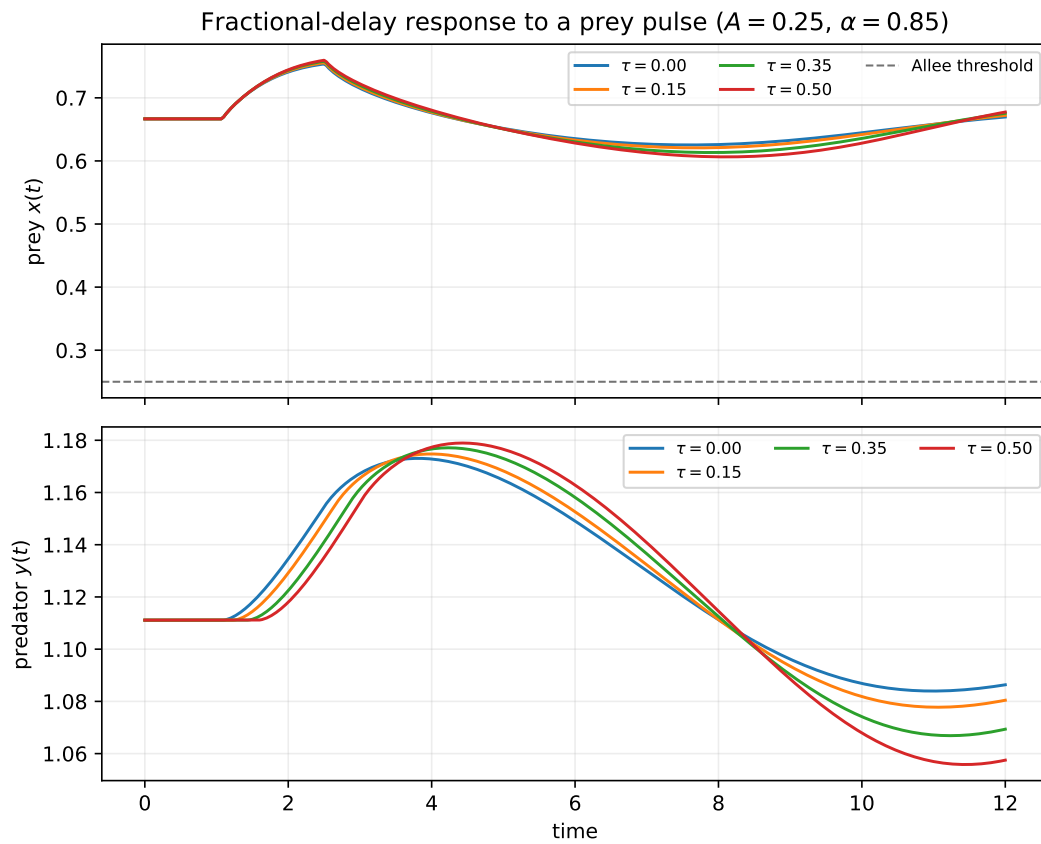


Figure 9. Combined fractional-delay response to the same prey pulse at $A = 0.25$ and $\alpha = 0.85$. Increasing delay shifts the predator response and reshapes the late transient while the fractional-memory relaxation pattern persists.

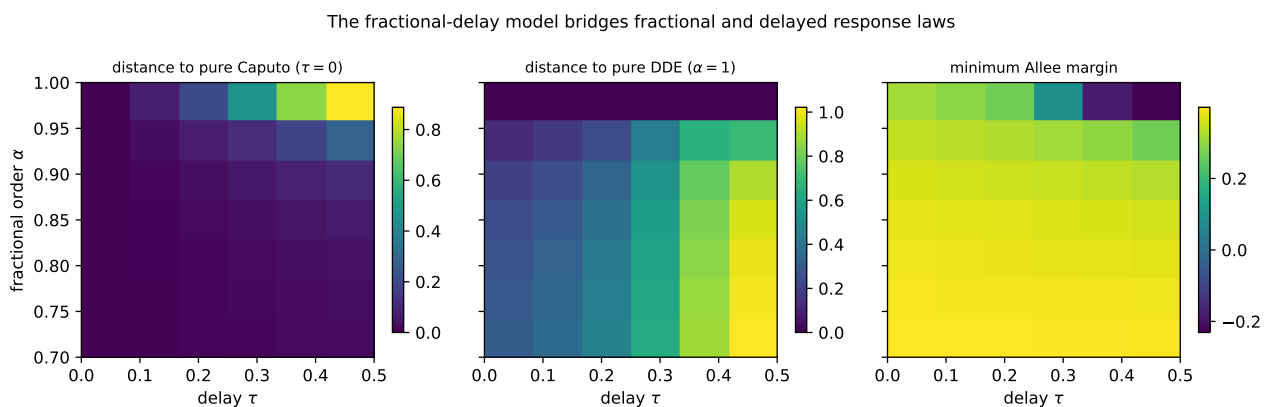


Figure 10. Fractional-delay bridge over (α, τ) . Left: trajectory distance to the pure Caputo limit. Center: distance to the pure DDE limit. Right: minimum Allee margin under the same pulse. Parts of the interior are poorly represented by either limit alone.

Table 6. Representative cells from the fractional-delay bridge at $A = 0.25$ under the pulse experiment. Distances are trajectory L^2 distances to the corresponding limiting models.

α	τ	distance to Caputo	distance to DDE	min. $[x(t) - A]$
0.75	0.10	0.0063	0.3426	0.3917
0.75	0.30	0.0182	0.6145	0.3887
0.75	0.50	0.0315	1.0086	0.3850
0.85	0.10	0.0103	0.2900	0.3724
0.85	0.30	0.0334	0.5683	0.3655
0.85	0.50	0.0625	0.9606	0.3566
0.95	0.10	0.0299	0.1645	0.3345
0.95	0.30	0.1142	0.4330	0.3123
0.95	0.50	0.2805	0.6875	0.2629

cross — and never discriminate particularly well either. The safety–informativeness tension survives intact on a model that contains both kinds of memory at once.

Table 7. Fractional-delay design stress test at $A = 0.25$. The score averages the minimum distance to the Caputo and DDE limits over $\alpha \in \{0.75, 0.85, 0.95\}$ and $\tau \in \{0.15, 0.30, 0.45\}$.

Input	mean bridge score	worst Allee margin	crossing fraction
multisine	0.1332	-0.2103	0.333
PRBS	0.0909	-0.2814	1.000
sinusoid	0.0894	-0.2777	1.000
chirp	0.0861	-0.2763	1.000
pulse	0.0627	0.2770	0.000
multiscale	0.0338	0.3171	0.000

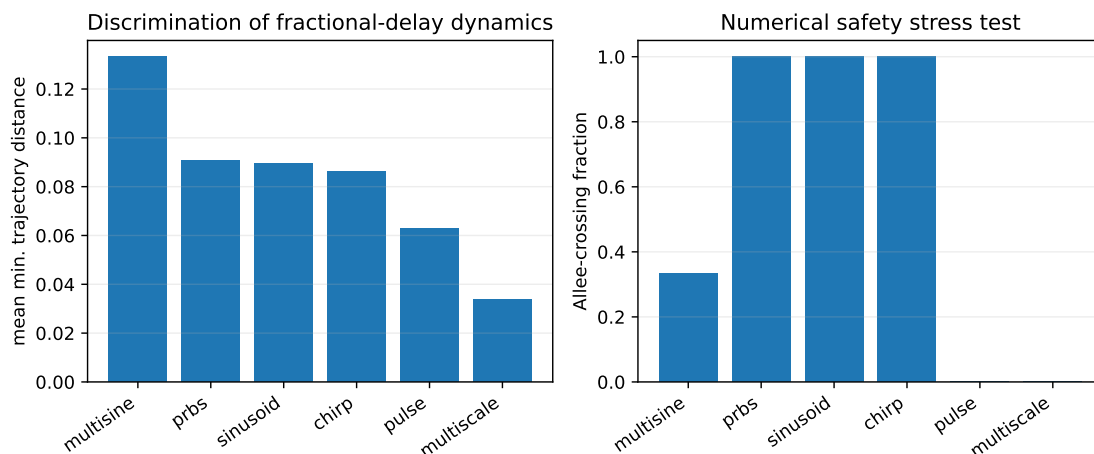


Figure 11. Input-design stress test for fractional-delay dynamics. Left: mean minimum trajectory distance from the two limiting explanations (pure Caputo and pure DDE). Right: numerical Allee-crossing fraction on the same (α, τ) grid. Broadband inputs expose the combined mechanism more strongly and are substantially less safe.

6.4. Delay sweep

The sweep in Figure 12 is where the mechanism becomes legible. Across every tested fractional order, more delay pushes the predator peak later and lifts its amplitude modestly, while the minimum Allee margin drifts downward — gently at low α , steeply near the integer-order limit. Delay, in other words, is not just one more fitting parameter; it announces itself as a reproducible shift in response timing, with a safety cost attached.

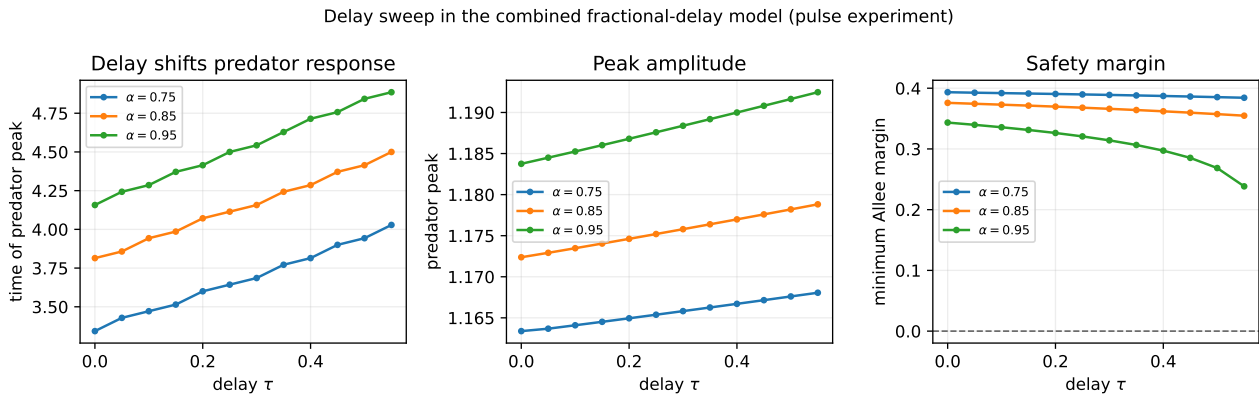
The numbers are worth recording. At $\alpha = 0.85$, moving from $\tau = 0$ to $\tau = 0.55$ shifts the predator peak from $t \approx 3.81$ to $t \approx 4.50$ and shaves the Allee margin from 0.3759 to 0.3548. At $\alpha = 0.95$ the same delay change moves the peak from $t \approx 4.16$ to 4.89 — and the margin collapses from 0.3434 to 0.2384, a far heavier toll. Table 8 gives the endpoints for all three orders.

These trends carry a practical lesson for sampling design. Delay speaks through timing; fractional memory speaks through a long relaxation tail. An observation scheme confined to a short window around the initial prey perturbation

Table 8. Endpoint summary of the delay sweep under the pulse experiment. Values compare $\tau = 0$ with $\tau = 0.55$.

α	τ	predator peak time	predator peak	min. Allee margin
0.75	0	3.343	1.1634	0.3934
0.75	0.55	4.029	1.1681	0.3843
0.85	0	3.814	1.1724	0.3759
0.85	0.55	4.500	1.1788	0.3548
0.95	0	4.157	1.1838	0.3434
0.95	0.55	4.886	1.1925	0.2384

risks missing both voices — too impatient for the delayed predator response, too brief for the fractional tail. Dense sampling around the predator peak, continued long enough to resolve the tail, listens to both.

**Figure 12.** Delay sweep under the pulse experiment. The predator peak arrives later as τ grows, its amplitude increases moderately, and the numerical Allee margin shrinks — consistently across the three displayed fractional orders.

7. Experiment Design and Safety

The design theory itself is classical in outline; we state just enough here to motivate the waveforms, and refer the full development with proofs to the Supplementary Material. For two Gaussian linearized models with common covariance, the pairwise information carried by an input u takes the quadratic form

$$D_{ij}(u) = \frac{1}{2} \langle u, Q_{ij} u \rangle,$$

so under an energy constraint the unconstrained optimum is the leading eigenfunction of Q_{ij} , and a robust frequency-domain design admits an optimum with finite frequency support. Neither fact is surprising; both are useful, because together they justify restricting attention to the six implementable waveform families of Figure 13 rather than searching an infinite-dimensional input space.

7.1. Strong-Allee safety

The one-sided Allee constraint demands

$$x(t) > A + \eta$$

throughout the experiment. For the linearized system, a sufficient pulse-amplitude envelope follows from the prey impulse gain; for the nonlinear system, the strict inward-pointing rectangle theorem and its four face inequalities are developed in the Supplementary Material. One subtlety deserves emphasis in the main text because it shapes every quantitative statement we make: state safety alone does *not* bound input energy. A stable, strictly proper channel attenuates high frequencies, so an adversarially oscillatory input can carry unbounded energy while the state barely moves. Any honest testing bound must therefore be anchored to an explicit actuator or protocol class — ours use the declared peak-amplitude pulse budget — and that is not a technicality but a structural feature of the problem.

8. Large-Scale Model-Discrimination Benchmark

The benchmark pits ODE, Caputo, retarded DDE, and finite latent mechanisms against one another at matched operating conditions. The primary stable-regime task lives at $A \in \{0.20, 0.25\}$; a separate task at $A > 2/7$ probes the

Six candidate designs at a common peak-amplitude budget $\|u\|_\infty = 0.10$, grouped by safety behavior

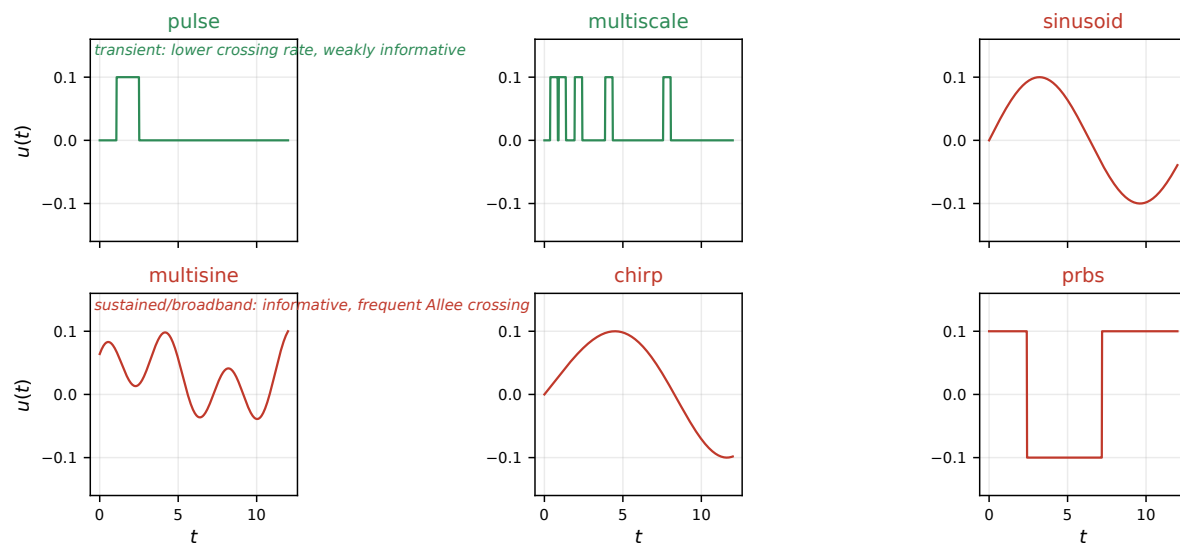


Figure 13. Six candidate perturbations at the common peak-amplitude budget. Transient and broadband designs interrogate the memory law in different ways — and behave very differently with respect to safety once the nonlinear simulations run.

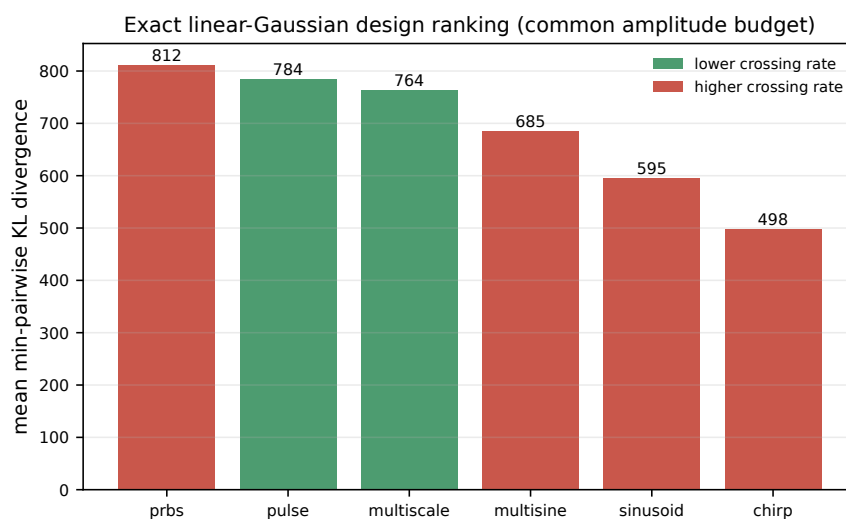


Figure 14. Linear-Gaussian design ranking by mean minimum pairwise KL divergence. Nonlinear state safety is invisible at this level, so the ranking serves as an information benchmark, not as a final experimental recommendation.

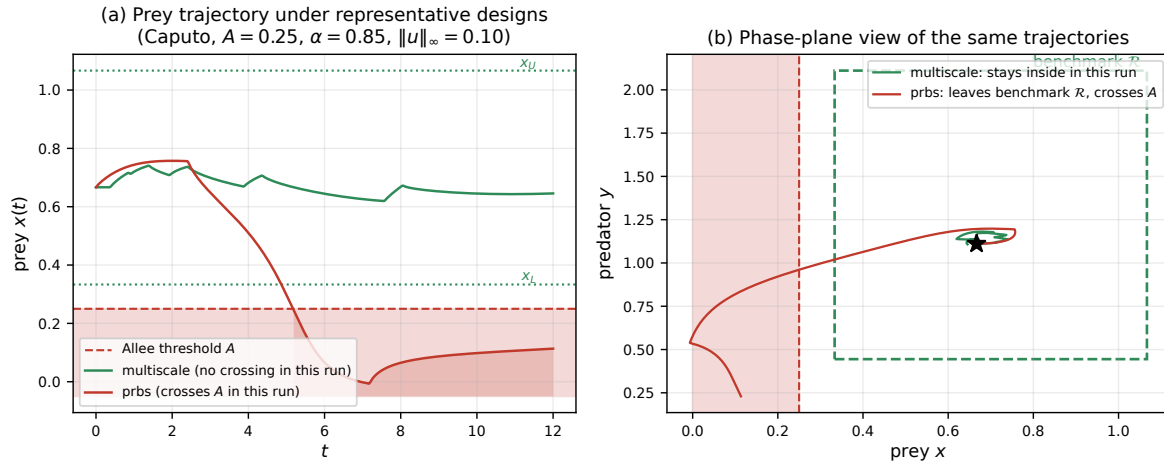


Figure 15. Representative nonlinear safety geometry. In the run shown, a multiscale transient stays above the Allee threshold while PRBS crosses it. The rectangle drawn is a benchmark diagnostic region; rigorous positive invariance requires the strict face conditions stated in the Supplementary Material.

easier stability-verdict question. Throughout, the decision rule is BIC — an information criterion, not Bayesian model evidence, and we never dress it up as the latter.

8.1. Benchmark design and decision rule

A single synthetic trajectory can flatter any method, so the benchmark is not built around one. The question it asks is sturdier: do the qualitative design conclusions survive as memory strength, observation noise, observation channel, and input waveform all vary? For each generated trajectory, every candidate model is fitted under the same observation protocol and the candidates are compared by BIC. What gets evaluated is a reproducible decision rule, nothing more.

Two factorial layers coexist and must not be conflated. The stable-regime nonlinear benchmark comprises 1512 cells with 200 stochastic replicates each and, for the record, zero solver divergences. A linear-Gaussian factorial of 1458 cells serves purely as an information-ranking reference. The separation matters because the two layers can disagree: the linear ranking says where models differ most locally, while the nonlinear layer knows about the strong-Allee geometry — and an input that tops the linear ranking can be disqualified outright once it starts crossing the threshold.

In the primary four-class stable task the macro-averaged accuracy is 0.537 against a chance level of 0.25. Ranking the input families:

$$\begin{aligned} \text{PRBS (0.584)} &\approx \text{multisine (0.583)} > \text{sinusoid (0.534)}, \\ &> \text{chirp (0.507)} > \text{pulse (0.414)} > \text{multiscale (0.324)}. \end{aligned}$$

Two axes dominate the difficulty. As $\alpha \rightarrow 1$ the mechanisms genuinely merge, and accuracy falls from roughly 0.70 at $\alpha = 0.70$ to 0.31 at $\alpha = 0.95$. Noise is the other master variable: 0.76 at high SNR, 0.44 at medium, 0.27 at low. Observing both species helps, though prey-only observation — the directly actuated channel — remains structurally sufficient. Table 9 collects the stratified numbers.

Table 9. Stable-regime four-class BIC accuracy stratified by fractional order, SNR, and observation channel.

Axis	level	accuracy	Axis	level	accuracy
α	0.70	0.702	SNR	high	0.756
	0.85	0.557		medium	0.444
	0.90	0.467		low	0.273
	0.95	0.312	channel	prey	0.516
				predator	0.427
				both	0.530

The class-wise statistics tell a story that a single accuracy number would hide. ODE recall is a striking 0.972 — but its precision is only 0.227, because whenever the excitation is too weak to reveal memory structure, the subtler mechanisms get absorbed into the simplest class. Caputo behaves in mirror image: recall 0.524, precision 0.984, so a Caputo verdict, once issued, is almost never wrong. DDE sits at 0.410/0.731 and latent-3 at 0.242/0.848. The

dominant failure mode is thus not chaotic confusion between mechanisms; it is a conservative collapse toward the simpler explanation when the data underdetermine the memory law.

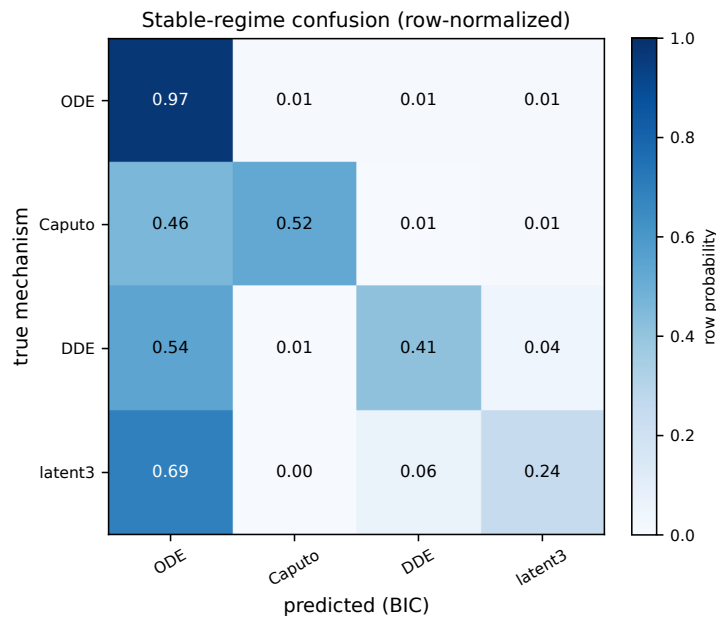


Figure 16. Row-normalized confusion matrix for the stable four-class task. Under weak excitation the simplest ODE mechanism absorbs the rest; Caputo predictions, when made, are highly precise.

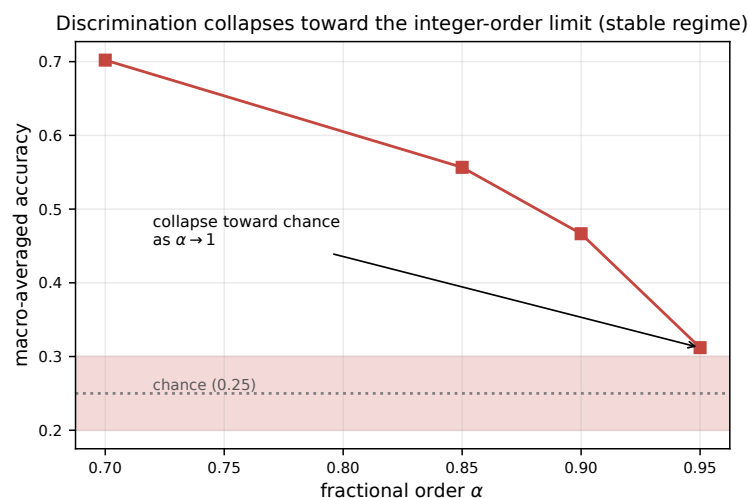


Figure 17. Model-selection accuracy versus fractional order. The closer α gets to the integer-order limit, the harder the memory mechanisms are to tell apart.

8.2. Safety–informativeness trade-off

We report numerical state safety on its own axis, never folded into the information score. At the common peak-amplitude budget the pattern is stark. Multiscale excitation never crosses the Allee threshold in the stable benchmark — and discriminates worst. PRBS and sinusoidal excitation discriminate best among the tested families and cross in essentially every stable-regime cell. **Multisine holds the interesting middle ground: discrimination on par with PRBS at a crossing rate of 0.357 rather than 1.0.**

Table 10 makes the frontier concrete. The safest family, multiscale, keeps a positive minimum margin of 0.292 at the price of the lowest accuracy. The pulse earns its keep as a compromise — a crossing rate of only 0.071, though its accuracy stays well below the broadband designs.

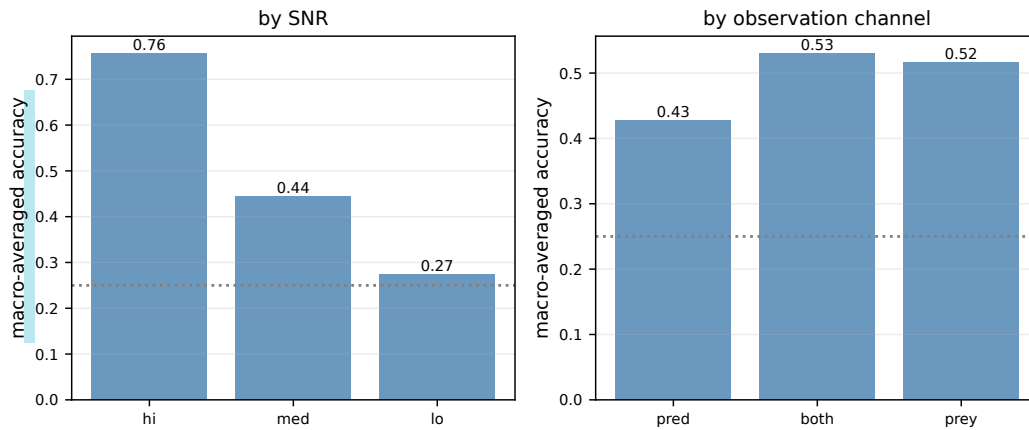


Figure 18. Accuracy stratified by SNR and observation channel. The benchmark is strongly noise-limited; observing both species gives the best average accuracy.

Table 10. Nonlinear stable-benchmark trade-off at the common peak-amplitude budget. Safety values are numerical benchmark diagnostics, not invariant-set certificates.

Input	macro accuracy	Allee crossing rate	minimum Allee margin
PRBS	0.584	1.000	-0.3139
multisine	0.583	0.357	-0.2405
sinusoid	0.534	1.000	-0.3058
chirp	0.507	0.929	-0.3012
pulse	0.414	0.071	-0.1207
multiscale	0.324	0.000	0.2921

The same table settles a methodological point: the linear-Gaussian ranking cannot pick the final experiment on its own. Linear theory judges an input by how well it separates response operators. The nonlinear ecology can respond to that very input by leaving the basin altogether — at which point there is no local identification experiment left to run. **Safety has to be read off the nonlinear state trajectory, not extrapolated from a linear information score.**

9. Prospective Experimental Interpretation

What would these results mean at the bench? Laboratory predator–prey microcosms offer exactly the kind of controlled perturbation setting our simulations presume [2, 7, 22], and the numerical evidence converts into fairly specific advice. A sensible first experiment operates in a stable strong-Allee regime, applies a bounded prey perturbation, watches both populations where the instrumentation allows, and holds the broadband artillery in reserve — deploying it only after a gentler probe has shown that the safety margin can absorb it.

The fractional-delay study contributes one diagnostic that is easy to state and easy to act on. Delayed feedback moves the *timing* of the predator peak; a lower fractional order reshapes the *long transient* and the phase. So an observation scheme that samples densely around the predator peak, and then keeps recording long enough to resolve the fractional tail, will see both signatures — something uniform sparse sampling cannot promise.

A Bayesian sequential layer is specified in the Supplementary Material but deliberately not executed: the evidence in the main paper stays within reach of deterministic solvers, interval enclosures, and BIC-based simulation benchmarks, all of which a reader can rerun.

A two-stage prospective protocol

Nothing in our results recommends firing the single most informative perturbation on day one. The frontier suggests a staged approach instead. Stage 1 applies a transient input with a low observed crossing rate — pulse or multiscale — to estimate the local response scale and confirm that the trajectory clears the Allee threshold with room to spare. Stage 2 is conditional: only when Stage 1 leaves adequate margin does the more informative multisine-type excitation come out. We make no claim that this sequencing is Bayes-optimal; it is a practical reading of the observed safety–information frontier, summarized in Table 11.

Three measurement priorities emerge. The directly actuated prey channel carries the strongest early signal and

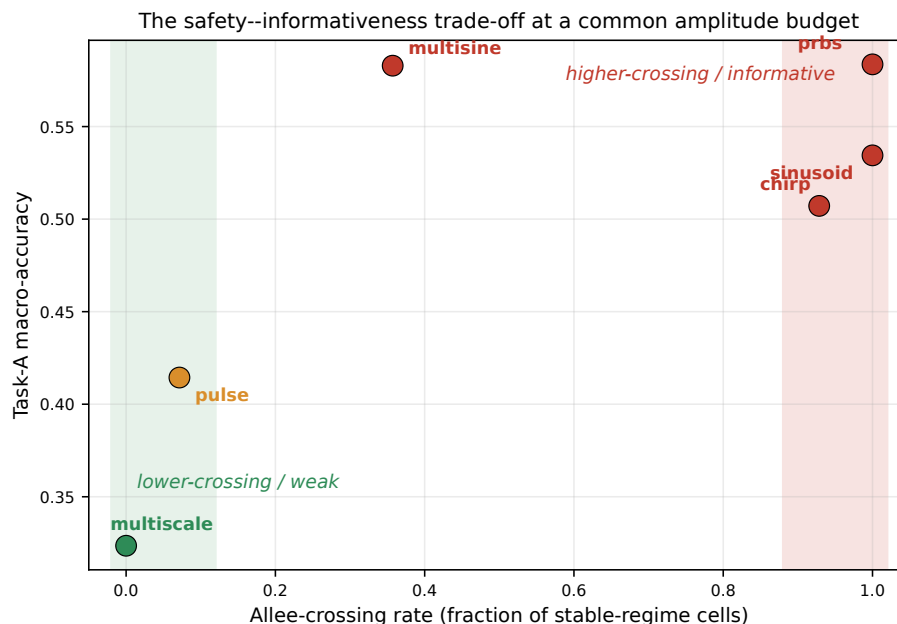


Figure 19. Headline benchmark trade-off. Broadband and sustained designs climb in discrimination but drift toward higher Allee-crossing rates; transient designs stay in the low-crossing, less-informative corner.

Table 11. Prospective microcosm workflow suggested by the simulation results.

Stage	Perturbation	Primary measurements	Decision / purpose
0	no perturbation	baseline prey/predator variability	estimate noise and confirm operating point
1	pulse or multiscale	dense early prey observations; predator peak timing; minimum prey level	verify local response and available safety margin
2	multisine if margin permits	both species over the full transient and relaxation tail	maximize separation of memory mechanisms
3	optional repeat / altered horizon	same channels, replicated under matched conditions	test robustness of the selected mechanism and estimate variability

deserves dense sampling right after the perturbation. The delayed trophic pathway speaks through predator-peak timing, so observations should crowd around the anticipated peak rather than spread thinly across the window. And fractional memory lives in the slow relaxation tail, which argues for a horizon extending well past the peak. Put together, the natural design is nonuniform: dense around the forcing and the predator peak, then sparser but longer through the relaxation.

A final caveat on units. The simulations run in dimensionless state variables under a normalized perturbation budget. Any physical implementation must first translate these into organism counts, biomass, or resource-manipulation units, and must set an operational safety margin above the *empirical* Allee threshold. That calibration is application-specific, and we resist the temptation to infer it from simulation alone.

10. Discussion

Three conclusions survive every layer of the study, analytical and numerical alike.

The first: *structural identifiability and practical discrimination are different questions*, and conflating them is the cardinal error this paper is built to avoid. Yes, the Caputo transfer carries an exact noninteger signature. And yes, finite latent models approximate the same response arbitrarily well on any finite horizon. The testing bounds put a number on the gap between those two facts — which is why a flexible latent rival stays troublesome long after the model classes have been proved analytically distinct.

The second concerns a habit of the literature: *treating delay and fractional memory as mutually exclusive explanations*. Our combined simulations trace a continuous bridge from the pure Caputo limit to the pure DDE limit, with delay governing the timing of the predator response and fractional order governing the broader relaxation structure. In between live dynamics that neither pure limit represents well. One cannot discover this by fitting the two endpoint models separately; it takes a direct numerical stress test of the composite mechanism, which is what Section 6 supplies.

The third is the one an experimentalist will feel most acutely: *the most informative perturbation is not the safest*, and the shortfall is not small. The four-class benchmark says so; the fractional-delay experiments say so again, independently. Broadband inputs amplify response differences and, in the same stroke, drive prey below the strong-Allee threshold more often. Nor is this a numerical artifact: crossing A rewires the basin structure, at which point the “experiment” has stopped being a local identification procedure and become an ecological intervention with consequences.

Limitations. We state these plainly. The certified ecological response theorem is linearized and specific to the prey channel; a full-state, hierarchy-wide safety closure remains open. The fractional-delay study is numerical, with a constant equilibrium history, a single delayed trophic pathway, and a deliberately targeted parameter grid. The benchmark decides by BIC, not by Bayesian evidence, and adaptive design is specified but never executed. Keeping this ledger explicit is what separates the proved results from the computational evidence.

Relation to the previously submitted companion paper. The shared material is the strong-Allee backbone and its locked baseline parameterization, both from Ref. [12] — and the sharing stops there. Ref. [12] is about certification: computer-assisted verification of Caputo equilibria, the Matignon stability boundary, extinction and boundedness, and the diagnostic failure of a forward-Euler surrogate. The present work is about discrimination. The structural transfer separation, the finite-horizon latent approximation hierarchy, the testing obstruction, the certified prey-response approximation, the six-family input-design study, the four-class benchmark, and the fractional-delay simulations appear in neither form nor substance in Ref. [12]. Where an inherited stability fact enters — always to select an admissible operating regime — it is attributed on the spot.

What is genuinely added by the present paper. Reusing the backbone is an experimental-design decision, not an act of recycling. With the ecology frozen, any difference between simulated observations must trace back to the memory law or the experiment — never to a quietly altered population model. The present paper thus begins where the companion paper ends. A certified operating point still leaves the hard inferential questions untouched: does an observed transient reveal a fractional mechanism uniquely? Can a delayed or finite latent model fake it? Which perturbations pry those alternatives apart without leaving the safe regime? Different questions, different mathematics, different computations.

The figures reflect the same division of labour. The companion paper’s centerpiece is a certified (A, α) stability atlas; ours are response separations, approximation errors, waveform designs, BIC confusion structure, fractional-

delay trajectories, and the safety–informativeness frontier. The shared model schematic exists to orient the reader in the common ecology — no certificate atlas and no certificate count from the companion paper is recycled as evidence here.

Relation to broader prior work. Safe information limits and safe active model discrimination exist as general theories [15, 19], and we make no claim to have invented safe experimental design. The contribution is the specific interaction: finite-horizon fractional approximation, bounded latent complexity, strong-Allee ecological geometry, and direct stress tests of the combined fractional-delay mechanism, all in one quantitative frame.

Implications for experimental identification. If there is a single moral, it is that model selection here is a *controlled intervention problem*, not passive curve fitting. Under weak excitation the four mechanisms produce trajectories close enough that BIC retreats to the simplest ODE explanation — conservatively, and often wrongly. Stronger excitation separates them and simultaneously courts the threshold. The fractional-delay study then adds its own warning: two endpoint models can each be individually identifiable while an interior composite occupies a response region neither represents, so designs should confront composite alternatives, not just the endpoints in turn.

The practical objective that emerges is not “maximize information” but maximize information *subject to* a trajectory-level safety margin and a declared rival-complexity budget. That second constraint is not bureaucracy: an unrestricted latent hierarchy forges fractional memory at will on any finite horizon, so the allowed latent dimension, delay range, forcing bandwidth, and observation horizon belong in the scientific specification of the experiment itself, alongside the organism and the protocol.

11. Conclusion

This paper set out to answer a deceptively simple question: what would it actually take, experimentally, to know that an ecological system remembers fractionally? The framework we built keeps the theory compact — four analytical results in the main text: structural separation, finite-horizon latent approximation, a complexity-dependent testing obstruction, and an interval-certified finite-state approximation of the prey response — and lets the numerical evidence carry the breadth: frequency-response diagnostics, a dedicated fractional-delay simulation study, six perturbation families with explicit safety metrics, and a large four-class discrimination benchmark.

Across all of it, one message repeats. Fractional memory is structurally distinct, yet a finite experiment may never separate it from a delayed or sufficiently rich latent alternative; pushing harder on the excitation buys separation at the price of Allee-threshold risk. The question worth asking before the next experiment, then, is not *which model fits best* — it is *which safe perturbation makes the competing memory mechanisms observably different*.

Author contributions

The author contributed to the conceptualization, methodology, formal analysis, numerical investigation, visualization, and preparation of the manuscript.

Use of AI tools declaration

AI-assisted tools were used during manuscript preparation for language editing, LaTeX formatting, code and documentation assistance, and structural organization. Responsibility for the scientific content and the final submitted version remains with the author.

Conflict of interest

The author declares no conflict of interest.

References

1. Elsayed Ahmed, A. M. A. El-Sayed, and Hala A. A. El-Saka. Equilibrium points, stability and numerical solutions of fractional-order predator–prey and rabies models. *Journal of Mathematical Analysis and Applications*, 325(1):542–553, 2007.

2. Lutz Becks, Frank M. Hilker, Horst Malchow, Klaus Jürgens, and Hartmut Arndt. Experimental demonstration of chaos in a microbial food web. *Nature*, 435:1226–1229, 2005.
3. Luděk Berec, Elena Angulo, and Franck Courchamp. Multiple allee effects and population management. *Trends in Ecology & Evolution*, 22(4):185–191, 2007.
4. Michele Caputo. Linear models of dissipation whose Q is almost frequency independent—ii. *Geophysical Journal International*, 13(5):529–539, 1967.
5. Franck Courchamp, Tim Clutton-Brock, and Bryan Grenfell. Inverse density dependence and the allee effect. *Trends in Ecology & Evolution*, 14(10):405–410, 1999.
6. Kai Diethelm. *The Analysis of Fractional Differential Equations: An Application-Oriented Exposition Using Differential Operators of Caputo Type*, volume 2004 of *Lecture Notes in Mathematics*. Springer, 2010.
7. Gregor F. Fussmann, Stephen P. Ellner, Kyle W. Shertzer, and Nelson G. Jr. Hairston. Crossing the hopf bifurcation in a live predator–prey system. *Science*, 290(5495):1358–1360, 2000.
8. Wiktor Jakowluk. Optimal input signal design for fractional-order system identification. *Bulletin of the Polish Academy of Sciences: Technical Sciences*, 67(1):37–44, 2019.
9. Wiktor Jakowluk and Mirosław Świercz. Optimal input design for fractional-order system identification using an LMI-based frequency error criterion. *Applied Sciences*, 15(23):12665, 2025.
10. Mohammad Javidi and Nasser Nyamoradi. Dynamic analysis of a fractional order prey–predator interaction with harvesting. *Applied Mathematical Modelling*, 37(20–21):8946–8956, 2013.
11. Anatoly A. Kilbas, Hari M. Srivastava, and Juan J. Trujillo. *Theory and Applications of Fractional Differential Equations*, volume 204 of *North-Holland Mathematics Studies*. Elsevier, 2006.
12. Maximiliano Lucius. Computer-assisted stability and extinction certificates for a caputo predator–prey system with a strong prey allee effect. Companion validated manuscript, draft dated 2026-07-30; published as Zenodo preprint DOI 10.5281/zenodo.21809908 (companion project fractional-memory-identifiability), 2026.
13. Rachid Malti, Mohamed Mayoufi, and Stéphane Victor. Experiment design for elementary fractional models. *Communications in Nonlinear Science and Numerical Simulation*, 110:106337, 2022.
14. Denis Matignon. Stability results for fractional differential equations with applications to control processing. In *Computational Engineering in Systems Applications*, volume 2, pages 963–968, Lille, France, 1996.
15. Xinpei Ni, Melkior Ornik, Glen Chou, and Samuel Coogan. Safe, real-time active model discrimination and fault diagnosis for nonlinear systems via differentiable reachability, 2026. arXiv:2606.19590.
16. Igor Podlubny. *Fractional Differential Equations*, volume 198 of *Mathematics in Science and Engineering*. Academic Press, 1999.
17. G. Ranjith Kumar and K. Ramesh. Dynamical analysis of fractional-order bazykin’s model with prey refuge, gestation delay and density-dependent mortality rate. *Iranian Journal of Science*, 49(1):79–91, 2024.
18. Fathalla A. Rihan. Fractional-order delay differential equations with predator–prey systems. In *Delay Differential Equations and Applications to Biology*. Springer Singapore, 2021.
19. Venkatesh Saligrama. When can safe controllers adapt? information before commitment, 2026. arXiv:2607.16895.
20. Àngela Sebastià Bargues, José-Luis Polo Sanz, Irene García-Camacha Gutiérrez, and Raúl Martín Martín. Practical implementation of optimal experimental design using the fractional-order fricke–morse bioimpedance model. *Chaos, Solitons & Fractals*, 170:113374, 2023.
21. Philip A. Stephens, William J. Sutherland, and Robert P. Freckleton. What is the allee effect? *Oikos*, 87(1):185–190, 1999.
22. Takehito Yoshida, Lee E. Jones, Stephen P. Ellner, Gregor F. Fussmann, and Nelson G. Jr. Hairston. Rapid evolution drives ecological dynamics in a predator–prey system. *Nature*, 424:303–306, 2003.